

[Biotin: Fact Sheet for Consumers \(original version\)](#)

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What is biotin and what does it do?

Biotin is a B-vitamin found in many foods. Biotin helps turn the carbohydrates, fats, and proteins in the food you eat into the energy you need.

How much biotin do I need?

The amount of biotin you need each day depends on your age. Average daily recommended amounts are listed below in micrograms (mcg).

Life Stage	Recommended Amount
Birth to 6 months	5 mcg
Infants 7–12 months	6 mcg
Children 1–3 years	8 mcg
Children 4–8 years	12 mcg
Children 9–13 years	20 mcg
Teens 14–18 years	25 mcg
Adults 19+ years	30 mcg
Pregnant teens and women	30 mcg
Breastfeeding teens and women	35 mcg

What foods provide biotin?

Many foods contain some biotin. You can get recommended amounts of biotin by eating a variety of foods, including the following:

- Meat, fish, eggs, and organ meats (such as liver)
- Seeds and nuts
- Certain vegetables (such as sweet potatoes, spinach, and broccoli)

What kinds of biotin dietary supplements are available?

Biotin is found in some multivitamin/mineral supplements, in B-complex supplements, and in supplements containing only biotin.

Am I getting enough biotin?

Most people get enough biotin from the foods they eat. However, certain groups of people are more likely than others to have trouble getting enough biotin:

- People with a rare genetic disorder called biotinidase deficiency
- People with alcohol dependence
- Pregnant and breastfeeding women

What happens if I don't get enough biotin?

Biotin deficiency is very rare in the United States. Biotin deficiency can cause thinning hair and loss of body hair; a rash around the eyes, nose, mouth, and anal area; pinkeye; high levels of acid in the blood and urine; seizures; skin infection; brittle nails; and nervous system disorders. Symptoms of biotin deficiency in infants include weak muscle tone, sluggishness, and delayed development.

What are some effects of biotin on health?

Scientists are studying biotin to understand how it affects health. Here is an example of what this research has shown.

Hair, nail, and skin health

Dietary supplements that contain biotin are often promoted to improve the health of your hair, skin, and nails, but there is little scientific evidence to support these claims. In a few small studies, some people with thin and brittle nails who took high doses of biotin had harder nails. Doctors have also reported that in a few cases, high doses of biotin have improved a rare hair disorder in children and skin rash in infants. More research is needed before biotin supplements can be recommended for any of these conditions.

Can biotin be harmful?

Biotin has not been shown to cause any harm. However, supplements that contain biotin above recommended amounts may cause false results in some lab tests, including those that measure levels of certain hormones, like thyroid hormone.

Does biotin interact with medications or other dietary supplements?

Yes, some medications you take may affect your biotin levels, and biotin may interact with certain medications. For example, treatment for at least 1 year with antiseizure medications (used to treat epilepsy) can significantly lower biotin levels.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if the dietary supplements might interact with your medicines or laboratory tests, or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients such as biotin.

Biotin and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about biotin?

- For general information about biotin
 - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on [Biotin](#)
- For more information on food sources of biotin
 - ODS Health Professional Fact Sheet on [Biotin](#)
- For more advice on choosing dietary supplements
 - ODS [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

alcohol dependence

A chronic disease (it lasts a person's lifetime) in which a person is unable to stop drinking once he or she has begun, needs to drink larger amounts of alcohol to get high, and suffers withdrawal symptoms (such as nausea, sweating, shakiness, and anxiety) after stopping drinking. The risk of developing alcohol dependence is influenced by a person's genes and lifestyle. Also called alcoholism.

biotin

A nutrient that is needed by the body to change carbohydrates, fats, and amino acids into energy and the basic materials needed for important life processes. It belongs to the group of vitamins called the vitamin B complex. Biotin is found in some foods, including egg yolk, liver, and yeast.

biotinidase deficiency

An inherited condition in which the body cannot use the biotin (a B vitamin) in foods. The body needs biotin to help break down food into energy. Symptoms of this condition typically appear in infancy or childhood. Kids with mild biotinidase deficiency may have weak muscles, skin rashes, and hair loss. Severe biotinidase deficiency can cause seizures, breathing problems, hearing and vision loss, problems with movement and balance, and an infection called candidiasis. Affected children also grow and develop more slowly. Biotinidase deficiency can be treated by a healthcare provider with high doses of biotin.

deficiency

An amount that is not enough; a shortage.

delayed development

Failure of a child to reach physical or behavioral milestones (such as rolling over, crawling, walking, and talking) at expected ages.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

disorder

In medicine, a disturbance of normal functioning of the mind or body. Disorders may be caused by genetic factors, disease, or trauma.

dose

The amount of medicine or other substance taken at one time or over a specific period of time.

epilepsy

A group of disorders that sometimes disrupts proper communication between brain cells, causing a seizure (a sudden change in behavior due to excessive electrical activity in the brain). It usually occurs in young children and the elderly. Epilepsy can be caused by abnormal brain development, brain damage, illness, tumors, or strokes. Often the cause is not known.

evidence

Information used to support the use of a particular screening procedure, treatment, or preventive measure. In medicine, evidence needed to determine effectiveness is provided by laboratory research, clinical trials, and other studies.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

genetic disorder

A disease or disorder caused by an alteration or variation (mutation) in a gene or group of genes in the cells of an individual. Examples of genetic disorders include breast cancer, cystic fibrosis, Parkinson's disease, and celiac disease. They can be inherited or can occur without a known cause.

hormone

A group of chemicals made by glands in the body. Hormones circulate in the bloodstream and control the actions of certain cells or organs. Some hormones can also be manufactured.

infant

A child younger than 12 months old.

infection

The invasion and spread of germs in the body. The germs may be bacteria, viruses, yeast, or fungi.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

laboratory test

A medical procedure that involves testing a sample of blood, urine, tissue, or other substance collected from the body. Tests can help determine a diagnosis, plan treatment, check to see whether treatment is working, or monitor a disease over time.

liver

A large organ located in the right upper abdomen. It stores nutrients that come from food, makes chemicals needed by the body, and breaks down some medicines and harmful substances so they can be removed from the body.

microgram

μg or mcg. A unit of weight in the metric system equal to one millionth of a gram. (A gram is approximately one-thirtieth of an ounce.)

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

nervous system

The brain and spinal cord, including the network of nerves that carry messages back and forth between the brain and all parts of the body. The nervous system controls what the body does.

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

pinkeye

An infection or inflammation of the lining of the eyelid and the white part of the eye. Also called conjunctivitis.

prescription

A written order from a health care provider for medicine, therapy, or tests.

protein

A molecule made up of amino acids that the body needs for good health. Proteins are the basis of body structures such as skin and muscle, and substances such as enzymes and antibodies.

seizure

Sudden changes in behavior caused by excessive electrical activity in the brain.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom

A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

thyroid

A gland located in the front of the neck, below the larynx (Adam's apple). The thyroid makes hormones that circulate in the bloodstream and affect brain development, metabolism, weight, breathing, heart rate, blood pressure, nervous system functions, body temperature, muscle strength, skin dryness, menstrual cycles, and cholesterol levels.

treat

To care for a patient with a disease by using medicine, surgery, or other approaches.

urine

Excess liquids and wastes that have been filtered from the blood by the kidneys, stored in the bladder, and removed from the body through the urethra (the tube that carries urine from the bladder to outside the body).

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

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